

IS 301: Data Warehousing (3 CU)

a) Description

This course covers techniques and software tools that can assist management that deals with large amounts of data in management and business decision making. The course covers the fundamental differences between databases and data warehouses, the techniques of developing data warehouses as well as manipulating them to generate business strategic decisions.

b) Objectives

- i. To provide students with broad coverage and examples about data warehouse techniques and trends underlying current and future development
- ii. To equip students with skills of developing data warehouses
- iii. To equip students with skills of maintaining existing data warehouses
- iv. To equip students with skills of manipulating data warehouses to generate information for business decision making
- v. Study best practices and case studies for using data warehousing applications, data warehousing enterprise platforms, and integrated data warehousing/Business Intelligence applications

c) Learning outcomes

By the end of the course, the student should be able to:

- i. Gain an understanding of basic data warehousing applications and techniques, and how data warehousing enables business intelligence capabilities that are used across many industries.
- ii. To combine and consolidate data from the various databases scattered throughout a company into a data warehouse.
- iii. Have acquired hands-on experience with key components of an integrated data warehousing and business intelligence system using a leading industry commercial application package.
- iv. Use data warehousing/BI applications to create enterprise business intelligence and analytics applications for solving real world problems.

d) Teaching and Learning Pattern

Teaching will be in form of class lectures, tutorials, lab demonstrations as well as class presentations.

e) Indicative Course Content

Topic	CH
Introduction to Business Intelligence , data warehouses and Corporate Information	3
Data warehouse architectures	3
Data warehouse lifecycle and project management	3
Introduction to Dimensional Modeling	3
Building dimensional models	6
Data extraction, transformation and loading(ETL)	6
Online analytical processing (OLAP)	3
Data mining	6
Data warehousing and the Web	6
Matching information to users	3

f) Assessment method

Assessment will be in form of tests and practical assignments (40%) and final written examination (60%)

a) Reading lists

- i. Kimball, R. and Ross, M., 2011. The data warehouse toolkit: the complete guide to dimensional modeling. John Wiley & Sons.
- ii. Krishnan, K., 2013. Data warehousing in the age of big data. Newnes.
- iii. Ponniah, P., 2011. Data warehousing fundamentals for IT professionals. John Wiley & Sons.